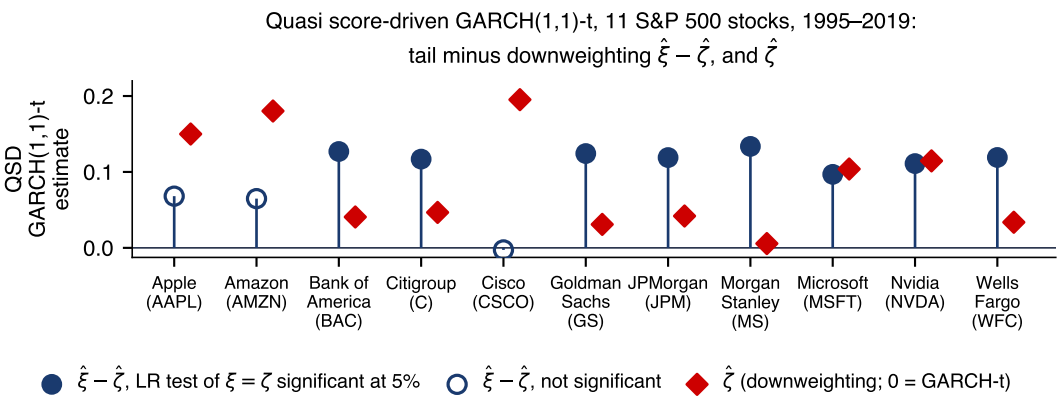
Quasi score-driven GARCH(1,1)-t, 11 S&P 500 stocks, 1995–2019:
tail minus downweighting $\hat{\xi} - \hat{\zeta}$, and $\hat{\zeta}$
QSD GARCH(1,1)-t estimate
0.2
0.1
0.0
Apple (AAPL)
Amazon (AMZN)
Bank of America (BAC)
Citigroup (C)
Cisco (CSCO)
Goldman Sachs (GS)
JPMorgan (JPM)
Morgan Stanley (MS)
Microsoft (MSFT)
Nvidia (NVDA)
Wells Fargo (WFC)
$\hat{\xi} - \hat{\zeta}$, LR test of $\xi = \zeta$ significant at 5%
$\hat{\xi} - \hat{\zeta}$, not significant
$\hat{\zeta}$ (downweighting; 0 = GARCH-t)